

East West University
Department of Computer Science & Engineering

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CSE366 : Artificial Intelligence
Section: 01

Assignment - 01

Submitted By

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Submitted To

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Question 1

Initial State S0: { on(b,a), ontable(a), ontable(c), ontable(d), clear(b), clear(c), clear(d), handempty }

Goal: { on(a,c), on(c,d) }

The 6-step plan being checked: Unstack(b,a) → Stack(b,d) → Pickup(c) → Stack(c,d) → Pickup(a) → Stack(a,c)

(a) Step-by-step trace — stopped at the first invalid action

Step	Action	Preconditions Check	State after action
S0 (before step 1)	—	—	on(b,a), ontable(a), ontable(c), ontable(d), clear(b), clear(c), clear(d), handempty
Step 1	Unstack(b, a)	on(b,a) ✓ clear(b) ✓ handempty ✓	ontable(a), ontable(c), ontable(d), clear(a), clear(c), clear(d), holding(b)
Step 2	Stack(b, d)	holding(b) ✓ clear(d) ✓	ontable(a), ontable(c), ontable(d), on(b,d), clear(a), clear(b), clear(c), handempty
Step 3	Pickup(c)	clear(c) ✓ ontable(c) ✓ handempty ✓	ontable(a), ontable(d), on(b,d), clear(a), clear(b), holding(c)
Step 4 (FAILS)	Stack(c, d)	holding(c) ✓ clear(d) ✗ FAIL	— action cannot run —

(b) Which step is wrong and why

Step 4 is the problem. Stack(c, d) needs clear(d) to be true before it can run. But look at what happened in step 2 — b was placed on top of d. That action removed clear(d) from the state. So from step 2 onwards, d has b sitting on it and is no longer clear. When step 4 tries to place c on d, the robot can't do it because the surface is taken. The precondition clear(d) simply isn't there in S3.

(c) Does skipping the wrong step allow the plan to reach the goal?

No. After step 3, the robot is holding c (hand is not empty). If step 4 is skipped, the next action in line is step 5: Pickup(a). That requires handempty — which is false, because the hand is still holding c. So step 5 also fails right away. The plan breaks down completely from step 4 onward, and the goal { on(a,c), on(c,d) } is never reached.

Question 2

Initial State S0: { ontable(a), ontable(b), ontable(c), clear(a), clear(b), clear(c), handempty }

Goal: { on(a,b), on(b,c) }

(a) Showing the 8-step plan is correct

Each state below shows what is true right after that action runs.

Step	Action	Preconditions Check	State after action
Step 1	Pickup(a)	clear(a) ✓ ontable(a) ✓ handempty ✓	ontable(b), ontable(c), clear(b), clear(c), holding(a)
Step 2	Stack(a, c)	holding(a) ✓ clear(c) ✓	ontable(b), ontable(c), on(a,c), clear(a), clear(b), handempty
Step 3	Unstack(a, c)	on(a,c) ✓ clear(a) ✓ handempty ✓	ontable(b), ontable(c), clear(b), clear(c), holding(a)
Step 4	Putdown(a)	holding(a) ✓	ontable(a), ontable(b), ontable(c), clear(a), clear(b), clear(c), handempty
Step 5	Pickup(b)	clear(b) ✓ ontable(b) ✓ handempty ✓	ontable(a), ontable(c), clear(a), clear(c), holding(b)
Step 6	Stack(b, c)	holding(b) ✓ clear(c) ✓	ontable(a), ontable(c), on(b,c), clear(a), clear(b), handempty
Step 7	Pickup(a)	clear(a) ✓ ontable(a) ✓ handempty ✓	ontable(c), on(b,c), clear(b), holding(a)
Step 8	Stack(a, b)	holding(a) ✓ clear(b) ✓	ontable(c), on(b,c), on(a,b), clear(a), handempty → GOAL REACHED ✓

Final state S8 contains on(a,b) and on(b,c), so both goal conditions are met. The plan works.

(b) Steps that are not needed

Steps 1, 2, 3, and 4 can all be removed. Here is why: compare S0 (the very start) with S4 (state right after step 4). They are the same — every fact that was true at the beginning is still true at S4, and nothing new was added. Steps 1 through 4 pick up block a, place it on c, remove it from c, and set it back on the table. Net change to the world: zero. The real work — building the stack — only starts at step 5. Dropping steps 1–4 still leaves a valid, complete sequence.

(c) Shortest possible plan

The same goal can be reached in **4 actions**:

Step	Action	Preconditions Check	State after action
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Step 1	Pickup(b)	clear(b) ✓ ontable(b) ✓ handempty ✓	ontable(a), ontable(c), clear(a), clear(c), holding(b)
Step 2	Stack(b, c)	holding(b) ✓ clear(c) ✓	ontable(a), ontable(c), on(b,c), clear(a), clear(b), handempty
Step 3	Pickup(a)	clear(a) ✓ ontable(a) ✓ handempty ✓	ontable(c), on(b,c), clear(b), holding(a)
Step 4	Stack(a, b)	holding(a) ✓ clear(b) ✓	ontable(c), on(b,c), on(a,b), clear(a), handempty → GOAL REACHED ✓

Why 4 is the floor and not 3 or fewer: the goal has two facts — on(b,c) and on(a,b). Getting any block onto another block takes at minimum two steps: one Pickup and one Stack. Two goal facts, two pairs of actions — that's 4 total, and there is no way around it with a single robot hand that holds one block at a time.